

Churn Prediction Proof-of-Concept: Recommendations Report

Prepared for: STADIOchoice Executive and Retention Teams

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Re: Recommendations arising from SS2 modelling results

This report summarises the results of our churn-prediction proof-of-concept, built using a comparable publicly available dataset (the PowerCo SME Customer Churn Dataset) while STADIOchoice's own subscriber data is finalised. It addresses which of the two candidate models is best suited to STADIOchoice's retention goals, how it can be improved, how it will need to change once real STADIOchoice data becomes available, and how these findings compare with existing published research.

1. Which Model Is Best?

Two models were tested and compared on identical, unseen data: a Logistic Regression baseline (Model 1) and a Random Forest ensemble (Model 2). Model 2 achieved substantially higher overall accuracy (86.5% vs 61.7%) and precision (28.9% vs 14.4%), meaning it wastes fewer retention offers on customers who were never actually at risk. However, Model 1 achieved more than double Model 2's recall (59.5% vs 26.8%) – it correctly identified far more of the customers who genuinely churned. A McNemar's test confirmed this difference in behaviour is statistically significant ($p < 0.000001$), and not simply due to chance.

For STADIOchoice specifically, recall is the more business-critical metric. The project's core motivation is catching subscribers before they cancel – including the “quiet leavers” who never contact support – so a missed churner (a false negative) represents a larger lost opportunity than one unnecessary retention offer (a false positive). On this basis, **Model 1 (Logistic Regression) is recommended**, despite its lower accuracy, because it more directly serves STADIOchoice's stated retention objective.

2. How Can the Selected Model Be Improved?

Model 1's main weakness is its low precision (14.4%), meaning most flagged customers do not actually churn. Three concrete improvements are recommended. First, adjust the model's decision threshold rather than using the default 0.5 cut-off: since Logistic Regression outputs a churn probability, raising this threshold slightly can improve precision without retraining, trading a small amount of recall for meaningfully fewer false alarms. Second, apply SMOTE (Synthetic Minority Oversampling) during training rather than relying on class-weighting alone; Al Mamun *et al.* (2026), using this identical dataset, achieved considerably higher accuracy using SMOTE alongside ensemble models, suggesting this

technique could meaningfully improve Model 1 as well. Third, expand feature engineering: the current engineered features (contract length, forecast accuracy, price change) meaningfully improved Model 2's predictions, with four of six ranking in its top ten most important features. Similar behavioural feature engineering – once STADIOchoice's real viewing and engagement data is available – is likely to sharpen Model 1's precision considerably.

3. Adapting the Model for STADIOchoice's Requested Data

The current proof-of-concept uses energy-sector features (electricity/gas consumption, pricing tariffs) as a stand-in for streaming behaviour. Once STADIOchoice's actual data (requested in the SS1 Data Request) is received, the model will need to be retrained, not simply re-applied. Energy-specific columns will be replaced with subscription and viewing-behaviour features: for example, “days since last login” and “search count” in place of consumption figures, and “retention offer accepted” in place of energy pricing variables. Reassuringly, the modelling pipeline, preprocessing steps, and evaluation methodology (IQR outlier capping, one-hot encoding, an 80/20 stratified split, and McNemar's test for model comparison) can all be reused directly, since none of these techniques are tied to a specific industry. One encouraging sign: STADIOchoice's real streaming churn rate (9.2%) is remarkably close to the PowerCo dataset's churn rate (9.72%), suggesting the imbalance-handling techniques validated in this proof-of-concept should transfer well to the real data.

4. Does This Align with the Related Literature?

These findings broadly align with the literature reviewed in Part A. Chang *et al.* (2024), studying telecom churn, also found Random Forest to be their best-performing model, achieving 86.94% accuracy – almost identical to Model 2's 86.5% here, despite being an entirely different dataset and industry. Mohan and Jadhav (2022), studying OTT/streaming churn directly, likewise found Random Forest outperformed simpler classifiers, reinforcing ensemble methods as a sound default choice for this type of problem.

However, one notable difference emerged: Al Mamun *et al.* (2026), using this exact PowerCo dataset, reported considerably higher accuracy (94%) than achieved in this project. The most likely explanation is their use of SMOTE for handling class imbalance, compared to this project's use of class-weighting alone – a gap directly addressed in the improvement recommendations above. Chang *et al.* (2024) also noted a lack of demographic data as a limitation of their own model, a limitation this project has already anticipated by explicitly requesting demographic data from STADIOchoice in the SS1 Data Request.

Conclusion

Model 1 (Logistic Regression) is recommended for its stronger alignment with STADIOchoice's retention goals. With decision-threshold tuning, SMOTE, and richer behavioural features once real subscriber data arrives, its current precision limitations can be meaningfully addressed.

References

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